

Jatan J. Pandya

New York, NY, USA | +1 (413)-362-6768 | jatanjay212@gmail.com | linkedin.com/in/jatanjay | jatanjay.github.io

EDUCATION

University of Massachusetts Amherst

Amherst, MA, USA

- Bachelor of Science, Computer Engineering
- Bachelor of Science, Computational Linguistics

09/2023

09/2023

SKILLS

- Languages: C, C++, Python 3, JavaScript
- Embedded Systems: Embedded Linux OS, Zephyr RTOS, Linux kernel/user-space driver development
- Full Stack Development: AWS (IoT Core, S3), Bootstrap, PostgreSQL
- Protocols & Communication: I2C, UART, CAN, SPI, WiFi, LoRaWAN
- Tools: git, gdb, Bash scripting, Hardware Debugging (Oscilloscopes, Logic Analyzers)

WORK EXPERIENCE

QuireTech Engineering Consultants LLC

Cresskill, NJ, USA

Embedded Firmware Engineer

09/2023 – Present

- Sole firmware engineer for a production battery-powered **LoRaWAN** device; architected low-level device software spanning SPI, GPIO, power domains, and peripheral **drivers**, deployed at scale in live environments.
- Architected and built a concurrent Zephyr **RTOS** system in **C**, using prioritized threads, message queues, mutexes, timers, and workqueues to coordinate radio, UI, storage, and peripheral subsystems.
- Developed custom **device drivers** for NFC module and E-paper controller, including register-level access, IRQ/BUSY handling, power sequencing, and safe concurrent access on a shared SPI bus.
- Enabled multi-variant and multi-region product support from a single firmware codebase via configuration-driven feature flags and modular subsystems such as regional LoRa parameters and feature-gated peripherals.
- Ported an embedded NFC driver from Zephyr to **Linux userspace** on Raspberry Pi using spidev and lgpio, building CLI and Flask-based tooling for ISO15693 tag provisioning and audit logging.
- Took the device from prototype to production, supporting **500+ deployed units** operating across **2 continents** and 6 countries in live environments (airports, retail, public facilities).
- Developed C firmware for a facial skin micro-needling therapy device, delivering a robust solution to production within 7 weeks of the prototyping phase.
- Engineered real-time, low-latency firmware using ASF SDK on a 32-bit ARM-Cortex-M microcontroller, managing buttons, rechargeable battery, motor, LEDs delivering seamless user experience (UX) to customers.

PROJECTS

RPi CM4-based Boat Monitor

Cresskill, NJ, USA

- Developed fully **automated provisioning** and QA pipeline for **200+** Raspberry Pi CM4 units, validating GPIO, LEDs, CAN bus, Wi-Fi/Ethernet, GPS, and RTC functionality using **Bash** and **Python** scripts.
- Managed large-scale golden image deployment over a 42-port Ethernet switch, streamlining Debian-based image provisioning and ensuring repeatable production-ready setups.
- Built and maintained automated system and peripheral tests, exposing core Embedded Linux concepts including device trees, kernel configuration, network stack setup, and service monitoring.
- Optimized provisioning and QA workflow to reduce human bottlenecks and accelerate time-to-deployment for distributed units.

CardVerse – A Magic: The Gathering Card Sorting Machine

Amherst, MA, USA

- Architected an award-winning automated **card inventory machine** (\$7K Innovation Challenge prize) using ROS2 and Python, integrating real-time computer vision, **ML**, and **mechatronics** for accurate card authentication and grading.
- Developed a multi-state image processing and pattern recognition pipeline on Nvidia Jetson and Raspberry Pi, achieving >97% defect detection accuracy and >99% authentication accuracy using YOLOv8 and OpenCV.